

Xiangnan Feng

Postdoctoral Fellow

Address: Complexity Science Hub,
Metternichgasse 8, 1030 Wien
Austria
E-mail: fengxiangnan@gmail.com
feng@csh.ac.at
Website: <https://xiangnanfeng.xyz>
Phone: +43 (0)67761406012

Research Fields

Computational Social Science, Complex Network, Machine Learning, Data Mining, Statistical Physics

Experience

Complexity Science Hub, Vienna, Austria Postdoctoral Fellow, Transforming Economies Supervisor: Frank Neffke (Group Leader)	2023 - Now
Max Planck Institute for Human Development, Berlin, Germany Postdoctoral Fellow, Future of Work Supervisors: Iyad Rahwan (Professor), Alex Rutherford (Senior Research Scientist)	2021 - 2022

Education

Beihang University, Beijing, China Ph.D. in Mathematics Supervisor: Zhiming Zheng (Academician of Chinese Academy of Sciences), Wei Wei (Associate Professor)	2014 - 2020
Beihang University, Beijing, China B.S. in Mathematics	2010 - 2014

Academic Topics

Future of Work

We study the labour market in a perspective of complexity for various topics in economics including skill evolution, automation, employment, et al.

Graph Neural Networks

We build new graph neural network models to solve network problems including link prediction, graph classification.

Network Theory

We use information theory to study fundamental problems in network like centrality, Epidemiology, et al.

Game Theory on Networks

We study game theory on networks with various dynamic strategies

Minimum Vertex Cover Problem

We study the network structure features in minimum vertex cover problem, one of the classical NP-hard problems.

Publications

IGA: Isomorphism-aware graph alignment for robust unsupervised anchor link prediction Qiming Yang, Wei Wei, Ruizhi Zhang, You Lv, Chao Li, Xiangnan Feng <i>Data Mining and Knowledge Discovery</i> 40, 111	2026
COLF: Disentangled forecasting for non-stationary time series via contrastive learning	2026

- Bowen Pang, Liyi Huang, **Xiangnan Feng**, Wei Wei
Pattern Recognition, 114808
- Who is using AI to code? Global diffusion and impact of generative AI 2026
Simone Daniotti, Johannes Wachs, **Xiangnan Feng**, Frank Neffke
Science 391 (6787), 831–835
- Efficient link prediction in continuous-time dynamic networks using optimal transmission and Metropolis Hastings sampling 2026
Ruizhi Zhang, Wei Wei, Qiming Yang, Zhenyu Shi, **Xiangnan Feng**, Zhiming Zheng
IEEE Transactions on Network Science and Engineering 13
- SCS: Subgraph contrastive supervised neural network for link prediction 2025
Qiming Yang, Wei Wei, Ruizhi Zhang, **Xiangnan Feng**
Information Sciences, 122482
- The building blocks of software work explain coding careers and language popularity 2025
Xiangnan Feng, Johannes Wachs, Simone Daniotti, Frank Neffke
<https://arxiv.org/abs/2504.03581>
- Evolutionary dynamic of the coupled awareness-epidemic networks with higher-order structure 2025
Xiangyu Meng, Wei Wei, **Xiangnan Feng**, Zhenyu Shi, Baifeng Li
Physica A: Statistical Mechanics and its Applications 660, 130210
- Probabilistic Time Series Forecasting via Diffusion Models under Multi-Scale Guidance 2024
Bowen Pang, Liyi Huang, **Xiangnan Feng**, Wei Wei
Available at SSRN 5071353
- Link prediction by continuous spatiotemporal representation via neural differential equations 2024
Liyi Huang, Bowen Pang, Qiming Yang, **Xiangnan Feng**, Wei Wei
Knowledge-Based Systems 292, 111619
- A representation-learning-based approach to predict stock price trend via dynamic spatiotemporal feature embedding 2023
Bowen Pang, Wei Wei, Xing Li, **Xiangnan Feng**, Chao Li
Engineering Applications of Artificial Intelligence, Volume 126, Part A, 2023, 106849, ISSN 0952-1976
- Graphical representation and hierarchical decomposition mechanism for vertex-cover solution space 2023
Wei Wei, **Xiangnan Feng**
Applied Mathematics and Computation, Volume 458, 2023, 128264, ISSN 0096-3003
- The dynamic resilience of urban labour networks 2023
Xiangnan Feng, Alex Rutherford
Royal Society Open Science 10(7), 230214
- Representation learning of enhanced graphs using random walk graph convolutional network 2023
Xing Li, Wei Wei, Ruizhi Zhang, Zhenyu Shi, Zhiming Zheng, **Xiangnan Feng**
ACM Transactions on Intelligent Systems and Technology 2023 Feb 10
- Abstract: Shaping and Predicting the Urban Labor Markets 2021
Xiangnan Feng, Manuel Cebrian, Alex Rutherford
the 10th International Conference on Complex Networks and Their Applications, Madrid, Spain
- Representation learning of graphs using graph convolutional multilayer networks based on Motifs 2021
Xing Li, Wei Wei, **Xiangnan Feng**, Xue Liu, Zhiming Zheng
Neurocomputing, 2021, ISSN 0925-2312
- Effects of dynamic-Win-Stay-Lose-Learn model with voluntary participation in social dilemma 2021
Zhenyu Shi, Wei Wei, **Xiangnan Feng**, Ruizhi Zhang, Zhiming Zheng
Chaos, Solitons & Fractals, Volume 151, 2021, 111269, ISSN 0960-0779
- Graph classification based on skeleton and component features 2021

Xue Liu, Wei Wei, Xiangnan Feng , Xiaobo Cao, Dan Sun <i>Knowledge-Based Systems</i> , Volume 228, 2021, 107301, ISSN 0950-7051	
Research of Motif-based similarity for link prediction problem Chao Li, Wei Wei, Xiangnan Feng , Jiaomin Liu <i>IEEE Access</i> , vol. 9, pp. 66636-66645, 2021	2021
Dynamic aspiration based on Win-Stay-Lose-Learn rule in spatial prisoner's dilemma game Zhenyu Shi, Wei Wei, Xiangnan Feng , Xing Li, Zhiming Zheng <i>Plos one</i> , 16(1), e0244814.	2021
Core influence mechanism on vertex-cover problem through leaf-removal-core breaking Xiangnan Feng , Wei Wei, Xing Li, Zhiming Zheng <i>Journal of Statistical Mechanics: Theory and Experiment</i> , 2019.7 (2019): 073401	2019
Research on centralities based on von Neumann entropy for motifs Xiangnan Feng , Wei Wei, Zhiming Zheng <i>2019 International Conference on Artificial Intelligence and Computing Science</i>	2019
Exploring the heterogeneity for node importance by von Neumann entropy Xiangnan Feng , Wei Wei, Renquan Zhang, Jiannan Wang, Ying Shi, Zhiming Zheng <i>Physica A: Statistical Mechanics and its Applications</i> , Volume 517, 1 March 2019, Pages 53-65	2018
Optimal stabilization of boolean networks through collective influence Jiannan Wang, Sen Pei, Wei Wei, Xiangnan Feng , Zhiming Zheng <i>Physical Review E</i> , 97, 032305 – Published 13 March 2018	2018
Correlation research of centralities on complex network by statistical learning Ying Shi, Wei Wei, Xiangnan Feng , Zhiming Zheng <i>2018 2nd International Conference on Artificial Intelligence and Software Engineering</i>	2018
Identifying influential vertices in boolean networks through dynamical voter rank Jiannan Wang, Xiangnan Feng , Zhilong Mi, Ziqiao Yin, Zhiming Zheng <i>2017 IEEE 2nd Information Technology, Networking, Electronic and Automation Control Conference</i>	2017
Combined fit to BESIII data on $e^+e^- \rightarrow h_c\pi^+\pi^-$ and $\chi_{c0}\omega$ Xiangnan Feng , Xuyang Gao, Chengping Shen <i>International Journal of Modern Physics A</i> , 30, 1550142 (2015)	2015
Optimization model for malfunction detection in automatic lathe Zhenfu Wang, Menglun Wang, Sen Chen, Xiangnan Feng <i>Modular Machine Tool & Automatic Manufacturing Technique</i> , 2015, ISSN: 1001-2265 CN: 21-1132/TG	2015
Photovoltaic hut design based on the greedy algorithm Zhenfu Wang, Menglun Wang, Sen Chen, Xiangnan Feng <i>Acta Energiae Solaris Sinica</i> , 2013 Vol. 34 (10): 1775-1780	2013

Conferences and Presentations

Workshop Talk: The Building Blocks of Software Work 10th Summer School on Data and Algorithms for ST&I Studies, Leuven	2025
Conference Talk: Learning from a Software Task Space International School and Conference on Network Science, Maastricht	2025
Symposium Talk: Learning from a Software Task Space , Vienna Complexity Science Hub Symposium, Vienna	2025
Invited Talk: Hierarchical Structure of Networks and Minimum Vertex-Cover Lancaster University Leipzig	2024

Conference Talk: The Dynamic Resilience of Urban Labour Networks Conference on Complex Systems 2022, Palma de Mallorca	2022
Symposium Talk: Prediction the Future Labour Markets , Berlin Center for Human and Machine Symposium, Berlin	2021
Conference Talk: Shaping and Predicting the Urban Labor Markets the 10th International Conference on Complex Networks and Their Applications, Madrid	2021

Grants & Awards

Gemma 2 Academic Program GCP Credit Award (\$10,000) , Google	2024 - 2025
EU Joint Research Center , Global hubs and local clusters in IT: a task-based analysis of the changing geography of software jobs	2025
Outstanding Graduate of BUAA	2021
Academic Excellence Foundation of BUAA , (\$3000, 85 out of 700)	2019

Activities

Reviewer for NetSci/NetSci-X	2024, 2025
Reading Group: Mastering Metrics , Complexity Science Hub Organize the reading group of book Mastering Metrics: The Path from Cause to Effect.	2023
Volunteer in NetSci 2023 , Vienna, Austria	2023
Seminar: Elements of Statistical Learning , Beihang University Organize the seminar of statistical learning as the group leader.	2017 - 2018
Overwatch Replay Analyzer (ORA) Develop the open-source software to extract a event timeline from computer game Overwatch video, which is used by professional Overwatch League E-Sports teams	2017 - 2018
Manager of Website: “Future Garden” , the Official BBS of Beihang University	2016 - 2020
BUAA University Tennis Team	2018, 2020
BUAA University Flag Football Team	2016
Internship in China Academy of Information and Communications Technology	2015
Teaching Assistant in Calculus , Beihang University	2014 - 2015

Academic References

Frank Neffke Transforming Economies Group Leader Complexity Science Hub Vienna (CSH) ✉ neffke@csb.ac.at	Alex Rutherford Senior Data Scientist UK Communications Regulator Ofcom ✉ alexisadams@gmail.com	Wei Wei Associate Professor School of Mathematical Sciences, Beihang University ✉ weiw@buaa.edu.cn
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